Reinforcement learning makes de novo enzyme designs recombinable, expanding their reachable sequence space

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Abstract

Science abstract target: ∼125 words. Written last, on purpose — it should compress the actual results rather than lead them. The structure below follows the three-move spine: frontier, fragments, PETase. Numbers in `[brackets]` are the ones that must be filled from measurement, not estimated.

- **Hook.** De novo design now produces enzyme backbones faster than they can be optimized. Optimization needs libraries that are large, diverse *and* target-specific; existing methods deliver at most two of the three.
- **Move 1 — the frontier.** GRPO fine-tuning of LigandMPNN against a structure-prediction reward moves the diversity/fidelity trade-off rather than sliding along it: on a public benchmark, **[3.98]**× the pass rate at matched sequence diversity.
- **Move 2 — the fragments.** Splitting designs at fixed boundaries and recombining across them turns $\sum_i |F_i|$ synthesized parts into $\prod_i |F_i|$ testable variants, and the recombinants retain **[XX]** % of the parental pass rate — a physically buildable library of **[10^X]** members.
- **Move 3 — the demonstration.** Three rounds of closed-loop optimization on a de novo PETase: a pilot library on a fluorogenic probe, a larger library on a PET-proximal substrate, and a third library designed from the accumulated sequence–activity data.
- **Headline.** The best construct is **[250]**× faster than the starting design and exceeds every characterized natural enzyme on the same probe.
- **Implication.** Function-aligned search plus manufacturing-aware library construction plus an experimental closed loop is a general recipe for de novo enzyme optimization.

[TODO: abstract — write once the fragment and PETase numbers land]

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Document index

Every source file in `paper/`, what it holds, and how far along it is. **drafted** means real prose exists; **scaffold** means planning notes only; **needs data** means the section cannot be written until data lands.

File	Contents	Status
Build		
`main.tex`	Compile order; \input's everything	drafted
`preamble.tex`	Packages, palette, `\plan` / `\todo` macros, draft-mode switch	drafted
`Makefile`	`make pdf`; SVG→PDF via Inkscape	drafted
`refs.bib`	Bibliography; all entries machine-retrieved	drafted
Front matter		
`abstract.tex`	∼125-word abstract, written last	scaffold
`intro.tex`	Four-paragraph introduction arc	scaffold
`related_work.tex`	Status quo: backbone generation, filtering, RL for inverse folding, library design, the gap	drafted
Results — the spine		
`results/frontier.tex`	Move 1: RL moves the diversity/fidelity frontier. Public benchmark, referee-reproducible	scaffold
`results/fragments.tex`	Move 2: split + recombine → combinatorial library. Computational half measured; physical half in `experimental.tex`	scaffold
`results/experimental.tex`	Move 3: that library ordered and assayed. 3,135 recombined PETase designs; zero-shot AF3 vs measured activity, and its failure to transfer to a second serine hydrolase	drafted
Methods		
`methods/overview.tex`	GRPO, fragment libraries, surrogate	scaffold
Back matter		
`conclusion.tex`	Four-paragraph discussion and outlook	scaffold
Not in this build		
`notebook.tex`	Separate document: the E1–E21 record	drafted
`experiments.tex`	Arms, selection rules, slate E1–E21, algorithm audit; **\input by notebook.tex, not by main.tex**	drafted
`library_notebook.tex`	Separate document: the library arc, L1–L5	drafted
`library_experiments.tex`	Fragment retention, recombination, granularity, the transfer gap; **\input by library_notebook.tex**	drafted
Archived — see `archive/README.md`		
`archive/*`	The heme and protease arcs, one superseded outline, and the *plan* for a three-round closed-loop PETase campaign that has no data. The PETase data that does exist is in `results/experimental.tex`	drafted

The paper and the record are two documents. `main.tex` builds the paper (`make pdf`); `notebook.tex` builds the experimental record (`make notebook`); `make all` builds both. The split

is deliberate: the notebook keeps pre-registrations next to the results that falsified them, which is what makes a retraction checkable and what a paper cannot carry and stay readable. A claim moves from the notebook to the paper by being rewritten as prose, never by \input.

Two companion files live in `paper/` but are deliberately *not* part of either build:

- `outline_computational.md` — working brainstorm for a computational-first version of the paper: landscape analysis, the three candidate framings, seven proposed experiments (E1–E7), figure plan, and risk register. Section 2 is the LaTeX rendering of its landscape half.
- `figures/palette.py` — the plotting counterpart to the `cleo*` colors defined in `preamble.tex`.

Draft mode. This document currently compiles with `\draftmodetrue` in `preamble.tex`, which renders all planning notes in gray and all `\todo` markers in red. Setting `\draftmodefalse` hides every one of them, leaving only submission prose — useful for checking how much real text actually exists.

1 Introduction

Science research articles do not use `\section` labels in the main narrative. The headings here exist so this draft has a usable table of contents; strip them at submission. Paragraph scaffolding is kept visible so collaborators can see the intended arc. Target: four paragraphs, ∼800 words.

P1 — The promise, and where it stalls

- RFdiffusion and its successors design enzyme backbones from scratch for chemistries with no natural template. Backbone generation is no longer the bottleneck.
- First-pass designs are catalytically weak — typically orders of magnitude below a useful enzyme. The bottleneck moved downstream, to optimization.
- **The framing this paper adopts:** optimization is limited by what you can physically *build and test*, not by what you can compute. That reframes the problem as library construction.

[TODO: p1 — de novo design succeeds at backbones, stalls at activity]

P2 — Three properties, and why nothing delivers all three

- A library that can rescue a weak de novo enzyme has to be simultaneously **large** (the active region is far away), **diverse** (it must cover directions, not just distance) and **target-specific** (every member must still fold and hold the active site). Name these three explicitly here — they are the spine of the whole paper and each results section closes one of them.
- Site-saturation mutagenesis is specific and buildable but strictly local; it cannot reach a variant 30 mutations away.
- Vanilla MPNN re-sampling is structurally informed but functionally blind. Low temperature returns one solution sampled repeatedly; high temperature returns diversity that does not fold. Section 4 measures exactly this trade-off.
- Scale alone is available — Weinstein et al. [1] reach $\sim 10^{16}$ designs by making the generative model *manufacturing-aware*, factorizing it to match what combinatorial DNA assembly can physically sample. **This is the closest prior art to Move 2 and must be cited as such, not as a foil.** The distinction to draw: they constrain the model to the chemistry from the outset and demonstrate on antibody repertoires, where a natural distribution supplies the objective; we train for catalytic geometry first and factorize afterwards, because for a de novo enzyme there is no natural distribution to imitate and the reward has to be built.
- Cross-reference Section 2 for the full treatment.

[TODO: p2 — the three properties; SSM, MPNN resampling, and the scale-only approaches. Cite weinstein2026manufacturing here as nearest prior art]

P3 — What we contribute

- **(i) A frontier shift, not a knob.** GRPO fine-tuning of an MPNN sequence designer against a structure-prediction reward, with a KL anchor to the base model that holds the output distribution open. Measured on a public benchmark against the full temperature ladder.

  The two halves of this paper fine-tune different base models and the text must not blur them: every AME benchmark run is LigandMPNN (`model_type: ligand_mpnn` in all 118 run configs), because the benchmark's targets carry a ligand in the backbone; the PETase campaign is ProteinMPNN. Name the specific model wherever a number is quoted and the generic "MPNN policy" only where the claim is about the method.
- **(ii) Combinatorial realization.** Fragment splitting and recombination convert the trained policy's output into a library that is exponentially larger than the number of DNA parts synthesized, while keeping the pass rate the policy bought.

- **(iii) A closed loop that actually closed.** Three rounds on a de novo PETase, each library designed using the measured activity of the last.
- Frame (i) and (ii) as *method* and (iii) as *demonstration*. Resist the temptation to present three co-equal applications — the earlier draft did, and it diluted all three.

[TODO: p3 — three contributions]

P4 — Headline numbers + roadmap

- Frontier: **[3.98]**× pass rate at matched diversity on the AME benchmark; U_{15} **[803]** vs **[607]** for the best baseline temperature.
- Fragments: **[N]** parts → **[10^X]** reachable variants at **[XX]** % retained pass rate.
- PETase: **[250]**× over the starting design; best variant ∼**[30]** mutations out; exceeds characterized natural enzymes on the same probe.
- One-sentence roadmap of the three results sections.

[TODO: p4 — headline results and roadmap]

2 Status quo: where the field stands and where the gap is

This section is written as real prose (not scaffold) because it is the part of the paper that does not depend on our own unfinished data. Every citation here was verified against Crossref, the arXiv API, or Europe PMC on 2026-08-04; the retrieval procedure is documented at the top of `refs.bib`. Quantitative claims attributed to other groups are taken verbatim from their abstracts.

2.1 Backbone generation is solved well enough that sequence design is now the bottleneck

Deep generative models now produce enzyme backbones for a wide range of chemistries. RFdiffusion established de novo design of protein structure and function by denoising diffusion over backbone coordinates [2]. RFdiffusion2 removed the requirement that catalytic geometry be specified at the residue level: it designs enzymes directly from functional-group geometries without specifying residue order or performing inverse rotamer generation, and generates scaffolds for all 41 active sites in a diverse in-silico benchmark, compared with 16 for previous methods [3]. Parallel efforts have produced designed serine hydrolases [4], metallohydrolases with designed metal sites [5], and enzymes built by catalytic motif scaffolding [6].

The rhetorical work of this subsection is to concede generously. These are strong papers and several share our senior author. The point is not that backbone generation is inadequate — it is that its success relocates the bottleneck downstream, to sequence design and to what happens after the first 96 designs come back weakly active.

Two features of this literature matter for what follows. First, the sequence-design step is treated as a solved subroutine: backbones are passed to ProteinMPNN [7] or LigandMPNN [8] sampled at low temperature, and the resulting sequences are filtered against structure-prediction and geometric criteria. Second, the experimental readout is a small, one-shot test set: RFdiffusion2 reports active candidates after testing fewer than 96 sequences per catalytic mechanism [3]. Both features are appropriate for demonstrating that a backbone generator works. Neither addresses what to do when the resulting enzymes are, as they almost always are, orders of magnitude less active than useful.

2.2 Filtering is not search

The prevailing sequence-design protocol is generate-then-filter. Sequences are sampled from a functionally blind inverse-folding prior, then scored against structure-prediction confidence, backbone recapitulation, catalytic-geometry and preorganization criteria — AlphaFold3 [9], Boltz [10], and ensemble methods such as PLACER, whose introduction substantially improved experimental success rates in serine hydrolase design [4].

This is the paper's central rhetorical move, so state it plainly and early, and make sure the phrasing survives review: a filter is a selection operator, not a search operator.

A filter cannot create a sequence that passes; it can only surface the sequences that a functionally blind prior happened to emit. This has two consequences that the field has largely left implicit. Sampling temperature is lowered to raise the pass rate, which means pass rate is purchased directly with diversity: ProteinMPNN's own report notes that sequence diversity can be increased considerably at higher temperature with only a small decrease in recovery [7], and the corresponding pass-rate cost is why production pipelines sit near $T = 0.1$. And because the survivors are drawn from a narrow neighborhood of the prior's mode, they are strongly correlated with one another — so a library's nominal size overstates the number of genuinely independent attempts it represents.

2.3 Reinforcement learning has reached inverse folding, but not libraries

Aligning generative models to objectives via reinforcement learning is now standard practice, and Group Relative Policy Optimization (GRPO) [11] in particular has proven to be a memory-efficient fit for the regime where rewards are computed by an external oracle rather than learned. This machinery has begun to reach protein sequence design. ProteinZero is an online RL framework for inverse folding that combines structural guidance from ESMFold with a self-derived $\Delta\Delta G$ predictor, adds an embedding-level diversity regularizer to mitigate mode collapse, and reports reducing design failure rates by 36–48% relative to ProteinMPNN, ESM-IF [12], and InstructPLM on the CATH-4.3 benchmark, with success rates above 90% across diverse folds [13]. Notably, its formulation balances three terms — multi-reward optimization, KL-divergence from a reference model, *and* diversity regularization — so the anchor and the diversity mechanism are separate components. Diversity-regularized direct preference optimization has been applied to peptide inverse folding with related motivation [14].

ProteinZero is the closest prior art to our mechanism and must be confronted in the first paragraph a reviewer reads on this topic, not buried. The differentiation is real but it is about the *objective*, not the optimizer: (i) general folds on CATH vs. enzyme active sites with ligands and catalytic geometry; (ii) sequence-level metrics — designability, recovery, stability — vs. a library-level objective; (iii) a diversity *regularizer* defended as anti-mode-collapse hygiene vs. diversity as a first-class design goal justified by screening economics; (iv) no physical library. If compute allows, running ProteinZero as a baseline arm is the strongest possible version of this argument. **[TODO: decide whether ProteinZero is runnable as a baseline]**

A separate line of work has independently arrived at the structural move our selection experiments test. Explorative Modeling factors the *training loop* rather than the generation procedure: it samples K candidate matches between model generations and data and trains on the best, so that predictions commit to modes rather than blurring them, and reports that scaling exploration acts as a third pretraining axis beyond parameters and data — improving FLOP efficiency 4.1× and sample efficiency 6.2×, and reaching 1.43 FID on ImageNet without guidance [15]. Oversampling candidates and training on a chosen subset is exactly the operation our E18 arms perform, and their observation that the mode-seeking variant "applies no pressure to cover every mode, so on its own it is

mode-seeking and can collapse" is an independent derivation of the failure we predict for pure quality-directed selection.

> Cite this as convergent evidence, not as prior art we are extending, and be precise about the disanalogy — a reviewer who knows the paper will check. Their selection criterion is the training objective itself (minimum reconstruction loss against real data), so every candidate can be scored before one is chosen. Ours cannot be: the objective requires a structure prediction, which is the expensive step and the entire reason to oversample. We therefore select on *proxies* — policy log-probability, sequence distance — before any reward is known. That is why a selection rule can lose to random in our setting, which a minimum-loss rule cannot, and it is the reason E18 carries a `random` control at matched oversample: their paper positions exploration as a scaling axis where more candidates inherently help, and reports no ablation separating selection quality from simply processing more samples.

Read together, these two papers and our own results converge on the same failure from three directions. ProteinZero introduces an embedding-level diversity regularizer specifically to mitigate mode collapse [13]; Explorative Modeling reports that its mode-seeking variant applies no pressure to cover every mode and can collapse on its own [15]; and we find that correcting the RL objective converts a policy that diffuses outward into one that converges onto a narrow solution, with positional entropy falling from ∼1.7 to 1.06. Collapse is not incidental to reward-aligned generative training — it is what the corrected objective does by default, and it is the reason a library-level method cannot simply inherit a sequence-level one.

> Where we differ is in what the anchor has to do. ProteinZero carries a KL term *and* a separate diversity regularizer; we carry only the KL term, and E13 shows it is sufficient on its own — sweeping its weight moves per-design coverage U_{15} from 506 to 878 monotonically, with total library coverage peaking in an interior plateau. That is a claim about parsimony, not about superiority: their regularizer operates in embedding space and ours does not, and we have not run the two against each other. State it as "one control suffices in our setting", which is defensible, rather than "the regularizer is unnecessary", which we have not tested.

What this line of work does not address is the library. Its objectives and benchmarks are defined per sequence: is *this* sequence designable, stable, recovered. The quantity that determines an experimental outcome is a property of the set — how many distinct, genuinely different candidates clear the bar, and how far apart they are.

2.4 Library design assumes a natural enzyme

A mature literature does optimize libraries, but from the opposite starting point. MODIFY learns from natural protein sequences to infer evolutionarily plausible mutations and predict enzyme fitness, and explicitly co-optimizes predicted fitness and sequence diversity of starting libraries, prioritizing high-fitness variants while ensuring broad sequence coverage [16]. Active Learning-assisted Directed Evolution (ALDE) uses uncertainty quantification over an epistatic five-residue active site and improved the yield of a non-native cyclopropanation product from 12% to 93% in three rounds of wet-lab experimentation [17]. Machine-learning-guided cell-free expression platforms scale the assay side, mapping fitness across large variant sets [18].

> MODIFY is the single most important citation for our diversity claim, because it proves the field already accepts that fitness and diversity should be co-optimized in a library. That is helpful — we are not arguing for diversity from scratch. What we are arguing is that every existing method for doing so is unavailable in the de novo regime.

Every one of these methods presupposes what a de novo enzyme lacks. MODIFY's fitness prior is evolutionary, learned from natural homologs; a de novo backbone has no homologs and no alignment. ALDE and active-learning methods require a measurable starting fitness to bootstrap from; de novo designs typically start at or near the detection limit. And both operate over a handful of positions,

whereas the mutations that rescue a de novo enzyme are, in our hands, tens of substitutions from the starting design — a combinatorial volume that site-saturation and few-position libraries cannot reach.

2.5 The gap

Nobody has addressed library design for de novo enzymes. The regime is defined by three simultaneous absences: no evolutionary prior, so MSA-based fitness models do not apply; no starting activity, so active-learning loops have nothing to condition on; and a target region of sequence space too far away for local mutagenesis. The field's current answer — low-temperature inverse folding plus hard filters — is not a search procedure at all, and it responds to the difficulty of the regime by narrowing the search precisely when it should be widening it.

Close on the thesis so the reader carries it into the results. Draft formulation, to be tightened once F2 exists: function-aligned online RL converts in-silico filters from a rejection step into a design objective, producing de novo enzyme libraries that simultaneously pass the community's own fidelity criteria at higher rates and span far more sequence space — increasing the number of independent experimental bets per unit of screening effort.

[TODO: gap paragraph — final phrasing of the thesis sentence, once the headline figure is built]

3 Results: function-aligned RL moves the diversity/fidelity frontier

Claim. On an established public benchmark, CLEO produces libraries with the sequence coverage of high-temperature LigandMPNN and the pass rate of low-temperature LigandMPNN — a combination no single temperature achieves.

Why this section leads. The PETase result rests on assays that take months and cannot be re-run by a referee. This one is fully computational, uses a published benchmark and a published deposit, and can be reproduced end to end. It buys the method credibility before any wet-lab claim is weighed.

Status: **scaffold**, and close to **drafted**. Both the CLEO arms and the full temperature ladder are measured. What remains: reconcile the two CLEO rows below (Block 3 quotes the corrected 200-step anchored arm; the notebook's E10 table still carries a superseded pre-fix peak window), rebuild the two PCA panels, fold in the E20 rescue result, and attach the run-to-run error bars from E21.

Block 1 — Setup and the unit of comparison

We evaluate on the atomized motif enforcement (AME) benchmark introduced with RFdiffusion2, comprising 41 catalytic sites scaffolded into 100 backbones each. The published protocol assigns 8 LigandMPNN sequences per backbone and predicts each with 5 Chai models; a prediction succeeds when the motif all-atom RMSD to the design is below 1.5 Å and no backbone atom clashes with the ligand.

The aggregation rule matters more than it looks and is easy to get wrong. We verified it against the deposit rather than assuming: the per-site number reported in `AME_benchmark_summary.csv` is the fraction of the 100 backbones with *at least one* passing prediction out of the 40, matching to zero error across all 41 sites and both methods. It is a best-of-40, not a mean over predictions. Scoring one prediction per sequence and comparing rates therefore understates a comparable method by roughly threefold — in the deposit, only 36 % of the sequences that pass on some Chai model pass on all five. Every number below is on their unit.

- Pilot backbones: three drawn from the `pass` class, at 32/40, 28/40 and 10/40 passing predictions in the deposit.

- Our predictor is AlphaFold3 rather than Chai; best-of-N uses 5 diffusion samples per sequence.
- **Caveat to resolve before submission.** Our implementation of the one predictor-free benchmark quantity, `ligand_dist_des_ncac_min`, does not reproduce the published values: median absolute error 0.4745 Å over 40 backbones, *zero* exact matches (`experiments/ame/calibration_ligand_dist.csv`; the pass/fail call still agrees on 37 of 40). Since it shares parsing and superposition code with the motif RMSD, the absolute values are not yet comparable to theirs. Until it is closed, only the backbone that reproduces (M0097) supports a head-to-head claim.
- **This is a metric-definition gap, not a reference-structure gap.** An earlier draft attributed it to our scoring against the idealized backbone where RFdiffusion2 scores against the unidealized, packed design. That explanation is now excluded by direct measurement. We recovered every candidate design structure from the authors' 177 GB benchmark archive — idealized, unidealized, LigandMPNN-packed, LigandMPNN input backbone — and all four give the same distance to within 0.01 Å while all four miss the published value by ∼0.47 Å. The authors' own `metrics_cache` confirms the mechanism could not have worked: idealization displaces atoms by 0.097 Å mean and 0.274 Å max, an order of magnitude too small. Atom-subset and motif-inclusion variants are likewise excluded.

[TODO: ame — reproduce `ligand_dist_des_ncac_min` exactly; needs `per_sequence_metrics.py`, which is not in the deposit (outputs only). Structure hypothesis falsified 2026-08-15 — do not retry it.]

Scope of this caveat. It bounds only whether our absolute numbers may be set beside the published ones. Every comparison the paper actually makes — CLEO against LigandMPNN, the temperature ladder, the KL sweep, the objective ablation — runs through a single pipeline with a single reference, and none of them is affected.

Block 2 — Low-temperature sampling returns one solution, sampled repeatedly

The incumbent protocol samples LigandMPNN at low temperature. At $n = 96$ on M0097 this yields a high success rate — 58.3 % of sequences pass — but the passing designs are a tight cluster: mean pairwise Hamming distance 40.6 out of $L = 180$, and 162 distinct substitutions among any 15 of them. The library is close to one solution measured many times, and most of the folding budget spent distinguishing its members buys little.

Raising the temperature removes the redundancy and most of the signal with it. At $T = 1.0$ the same backbone reaches mean pairwise distance 126.3 at a success rate of 4.2 %. Diversity alone is not the objective; a library that cannot be filtered is not a library.

Two numbers here were wrong in earlier drafts and are corrected. $T = 1.0$ was reported as *zero* passing designs; that came from $n = 8$, and at $n = 96$ it passes 4.2 % of the time. And the low-temperature arm was described as collapsing to a single cluster "at every threshold down to 50 % identity" — true at 70 %, but at 90 % every passing design is its own cluster and the statistic simply counts designs. Cluster counts are threshold-dependent and are no longer quoted; mean pairwise Hamming and the distinct-substitution count U are threshold-free and replace them throughout.

Block 3 — cleo occupies the corner neither temperature reaches

Sampling temperature traces a frontier: success falls as diversity rises, and every point on it is available to the incumbent by turning one knob. The claim of this section is that CLEO sits above that frontier, and the measurement is in Table 1.

Interpolating the baseline frontier at the anchored arm's diversity, the incumbent succeeds 14 % of the time where CLEO succeeds 56.7 % — **3.98**×. Read the other way, at CLEO's success rate the incumbent reaches mean pairwise distance 59 against CLEO's 114.4 — **1.93**×. The anchored arm also carries more distinct substitutions per passing design than any baseline ($U_{15} = 803$ against 607 at T=0.7), so this is a genuine exploration advantage and not only a higher yield.

Table 1: M0097, per-sequence success rate on best-of-5 AF3 predictions, with diversity measured over the *passing* designs only — a library contains only what survives filtering. U_{15} is the number of distinct (position, residue) substitutions in a library of 15 passing designs, rarefied over 300 draws so arms with different yields are comparable. Baselines are $n = 96$; CLEO arms are the final 40 steps of a 200-step run.

Arm	**Success**	**Mean pairwise Hamming**	U_{15}	**Median RMSD** (Å)
LigandMPNN T=0.1	58.3 %	40.6	161	1.42
LigandMPNN T=0.2	66.7 %	47.8	205	1.37
LigandMPNN T=0.3	58.3 %	57.9	273	1.31
LigandMPNN T=0.5	34.4 %	80.5	437	1.70
LigandMPNN T=0.7	26.0 %	100.5	607	1.88
LigandMPNN T=1.0	4.2 %	126.3	—	5.59
CLEO, no anchor	83.1 %	76.6	506	1.07
cleo, KL anchor	56.7 %	**114.4**	**803**	1.44

The unanchored arm is the instructive contrast. It reaches the highest success rate in the table, 83.1 %, and the *worst* per-design coverage of any CLEO arm — $U_{15} = 506$, below T=0.7. Left unconstrained the policy converges onto a narrow solution. The KL weight against the base model is what holds the distribution open, and it is the control the library application needs.

Sweeping that weight over $\{0, 0.005, 0.01, 0.02, 0.05, 0.1\}$ shows it is a well-behaved control with an interior optimum rather than a knob that has to be tuned. Success falls monotonically, 83 % to 10 %, and per-design coverage rises monotonically, U_{15} 506 to 878; the total distinct substitutions a library actually contains is the product of the two and is flat within noise across $w \in [0.01, 0.05]$ (2480 to 2501) before collapsing to 1731 at $w = 0.1$, where too few designs pass for per-design exploration to accumulate. The value used throughout, $w = 0.02$, sits in the middle of that plateau, so the reported numbers do not depend on having found a sharp optimum.

Figure 1 shows why the success rate alone is not enough: on M0097 every CLEO arm has its *median* below the 1.5 Å cutoff, so the mode has moved rather than the tail merely reaching across — which is precisely what fails to hold on the near-miss backbones of Block 6.

Block 4 — The training trajectory is the library

This is the reframing that removes a hyperparameter. Rather than adding an explicit diversity term to the reward, we treat every rollout sampled during training as a library candidate. Those sequences were already folded and scored to compute the reward, so the candidate pool is free: the library is a byproduct of training, not an additional expense.

Across 200 GRPO steps at 16 sequences per step, M0097 accumulates 3200 scored designs. Under the anchored policy 1349 of them are distinct passing sequences carrying 3045 distinct substitutions — 96 % of the reachable ceiling. That ceiling is $18 \times 176 = 3168$, not $19L = 3420$: four catalytic positions are held fixed and cysteine is barred from every sampled position, so each of the 176 free positions offers 18 alternatives to its reference rather than 19. The passing set varies at exactly those 176 positions, which is the check that the denominator is the right one.

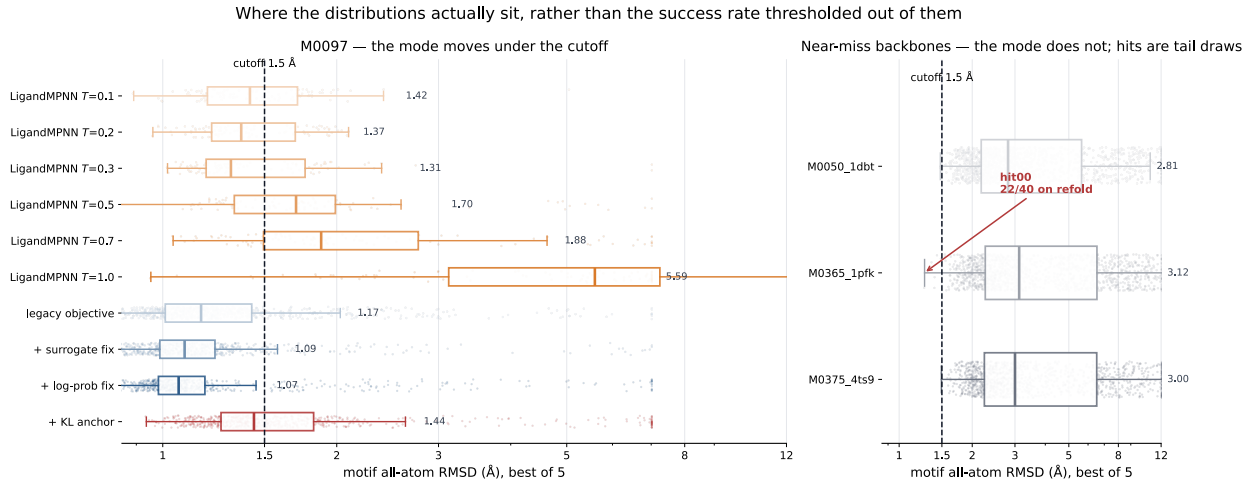


Figure 1: **Where the distributions actually sit, rather than the success rate thresholded out of them.** Each row is one arm's per-design best-of-5 motif RMSD: box for median and quartiles, jittered strip behind it for the shape the box hides. Log axis, because a linear one lets the single worst arm compress everything that matters into the left fifth. **Left, M0097:** every CLEO arm has its *median* below the cutoff (1.07 Å to 1.44 Å) — the mode has moved, not just the tail — against 1.31 Å to 1.42 Å for low-temperature LigandMPNN and 5.59 Å at T=1.0. The non-monotonicity noted in E10 is visible directly: T=0.3 has a *lower* median than T=0.1. **Right, the three near-miss backbones:** medians of 2.81 Å to 3.12 Å with the cutoff far out in the left tail. The one design that survived 40 refolds sits at the extreme edge of its own distribution. Encoding: both ladders are ordinal, so each is a single hue ramped light to dark rather than arbitrary categorical colours; the anchored arm keeps an accent hue as the highlighted result.

	cleo, no anchor		cleo, KL anchor	
Cumulative designs	**Unique passing**	**Substitutions**	**Unique passing**	**Substitutions**
400	75	1483	47	1382
800	284	2294	160	2045
1600	893	2805	510	2757
2400	1560	2974	903	2958
3200	2229	3040	1349	3045

The distinct-substitution count is genuinely saturating here, at 89 % to 89 % of the ceiling, while the count of distinct passing sequences is still climbing steeply. The library is not running out of designs; it is running out of *single* substitutions to explore, which is the expected behaviour once most positions have been visited and further diversity must come from combinations.

This supersedes an earlier version of this table reporting 97 passing designs from 3200 folds, a factor of 14 to 23 lower. That run used the pre-audit objective, and its curve flattened because the policy decayed rather than because sequence space was exhausted — the earlier draft said so and predicted rank normalisation would keep it climbing, which is what happened. The pool numbers here are best-of-5, matching Block 3; the superseded ones were single-sample and were not comparable to it.

Block 4b — Where in sequence space the passing designs sit

Generated by `paper/figures/ame_pca.py`, which writes two figures because one projection cannot honestly carry both claims. The shared projection is fit across arms within a backbone — a per-arm PCA places every cloud at the origin and destroys the comparison — but deliberately not across backbones, where a global fit puts 99 % of PC1 variance on backbone identity.

These two panels are stale and must be regenerated. They were built from the pre-audit runs and from baseline arms holding 8 sequences each. Both inputs have been replaced: the objective is fixed and the baselines are now $n = 96$. The drift control below remains the right control and its conclusion is unlikely to change, but no number in these captions should be quoted until they are rebuilt.

Explained variance is low (PC1 5 % to 9 %), as expected for one-hot protein sequence, so the projection is a visual aid; every quantitative diversity claim in this section uses Hamming distance and substitution counts directly.

[TODO: ame — rebuild both PCA panels on the corrected runs at matched n]

Block 5 — The temperature frontier

Measured, and reported in Table 1. Two predictions stated in advance were tested.

The dangerous intermediate temperatures do not close the gap. $T = 0.2$ and $T = 0.3$ were named as the points that would refute a frontier shift if either reached CLEO's coverage at CLEO's success rate. Neither does: both sit below 0.6 of the anchored arm's diversity. The frontier shift stands.

One prediction was wrong. The trade-off was expected to be monotone in temperature; it is not. $T = 0.2$ is the *most* successful arm in the table at 66.7 %, above $T = 0.1$'s 58.3 %, while also being more diverse. $T = 0.2$, not $T = 0.1$, is therefore the incumbent to quote at the low-diversity end, and any comparison against $T = 0.1$ alone flatters us.

Still outstanding from this block's original plan: the saturation fold count for low-temperature LigandMPNN — the point past which it re-emits sequences it has effectively already produced. The CLEO side of that comparison is Block 4; the baseline side needs a cumulative-distinct curve at matched budget, which the $n = 96$ sweep can supply but does not yet.

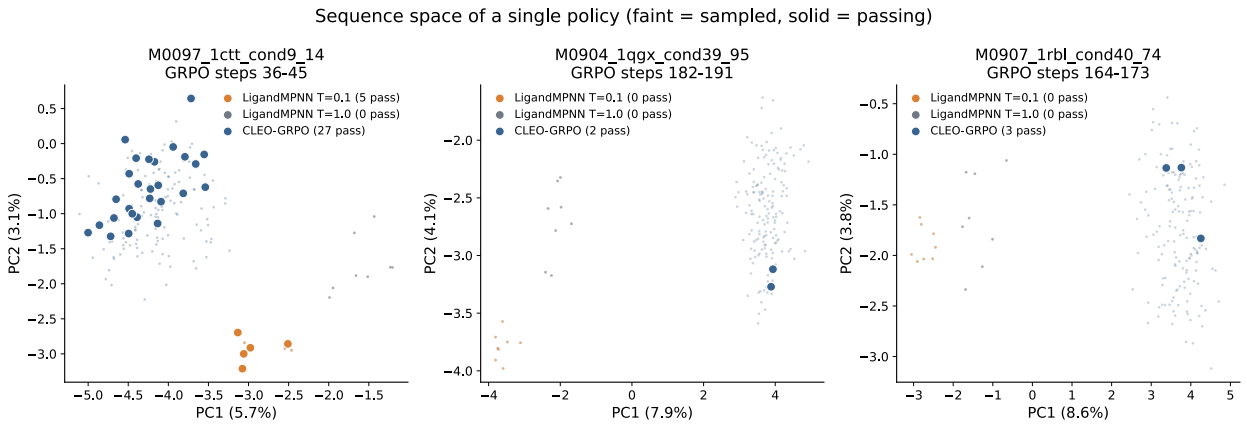


Figure 2: **Sequence space occupied by each design method on the AME pilot backbones.** One PCA per backbone, fit on every sequence that backbone contributes across all three arms, so the arms share one projection and the clouds are directly comparable. Faint points are sampled designs, solid points are designs that pass the benchmark criterion (motif all-atom RMSD < 1.5 Å, no ligand clash). CLEO is restricted to its peak-performing 10-step window so that three *policies* are compared at one point in training. On M0097 the low-temperature arm (orange) forms a tight knot displaced far along PC1 — five passing designs that are a single cluster at every identity threshold down to 50 % — while CLEO's passing designs are spread throughout a broad cloud. T=1.0 (grey) reaches comparable spread with zero passing designs. Projections are deliberately *not* shared across backbones: a global fit places 99 % of PC1 variance on backbone identity, collapsing each backbone to its own blob and hiding the arm differences entirely. Explained variance is low (PC1 5 % to 9 %), as expected for one-hot encoded protein sequence; the projection is a visual aid and every quantitative claim in the text uses Hamming distance and cluster counts directly. Baseline arms hold 8 sequences each, so part of their compactness is small-n.

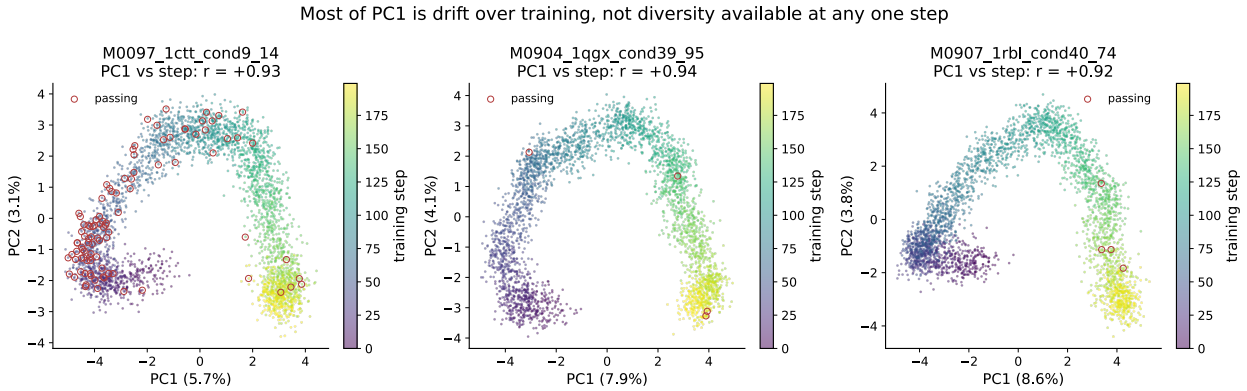


Figure 3: **Most of PC1 is drift over training, not diversity available at any one step.** The full GRPO trajectory for each backbone, in the same projection as Figure 2, coloured by training step; passing designs are circled. PC1 correlates with training step at $r = +0.92$ to $+0.94$ on all three backbones. This distinction is load-bearing rather than cosmetic: spread accumulated by drifting across training is genuine library diversity when the library is assembled from the whole run, which is how ours is assembled, but it is *not* evidence that any single policy is diverse. Presenting the pooled projection alone would conflate the two and overstate the result, so the axis is shown rather than hidden.

Block 6 — Rescue of backbones the benchmark cannot solve

Three backbones sit at 1.61 Å to 1.69 Å best-of-40 in the deposit — zero passing predictions, but only ∼0.15 Å from threshold. A pass on any of them is a design the published pipeline cannot produce, with no reproduction confound, because there is no positive result of theirs for us to fail to reproduce.

All three were trained for 75 steps at batch 16. Nine designs cleared the criterion. **Each was then refolded 40 times — the published per-backbone budget — and one survives.** M0365 `hit00` passes 22 of 40 predictions with a *median* RMSD of 1.470 Å, below the cutoff rather than touching it, on a backbone the deposit scores 0/40 with a best of 1.69 Å. That is a design the published pipeline does not produce.

The other eight do not survive. Five pass zero of 40; three pass one or two. Training-time margin predicts this almost perfectly: the single design with a margin above 0.03 Å inside the cutoff is the single design that replicates, and everything at 0.003 Å to 0.023 Å was a tail draw from a distribution whose median sits near 3 Å (Figure 1, right).

The honest claim is one rescue, not nine, and the general lesson is worth more than the rescue: a best-of-N filter applied to a distribution whose mode is far from the cutoff selects excursions, not solutions. A design that truly passes 1 in 40 times is flagged by a best-of-5 filter about 12 % of the time it is folded. Every headline number that leaves this section should be replicated the way `hit00` was, and in-training pass rates should be described as counts of best-of-N events rather than of reproducible designs.

Two caveats remain open. The fold budget is not matched — 1200 designs at best-of-5 against the deposit's 8 sequences at 5 models — so a compute-matched comparison is still owed. And the idealised-versus-unidealised reference structure question from Block 1 applies here as it does everywhere.

Block 7 — The cost axis: evaluations, not steps

Every number in this paper is bought with structure predictions, so the unit that matters is folds, not training steps. The two are not interchangeable: folds per step is rotations × batch × diffusion samples, so at fixed step count a batch-64 run spends eight times what a batch-8 run does. Comparing policies at matched steps compares budgets, not methods.

A batch sweep with sample reuse pinned at 4×, so that batch size is the only factor varying between arms, gives Figure 4. **Batch 16 reaches the** 90 % **chimera pass level on** 40 000 **evaluations against batch 32's** 55 000**.**

The arms are matched in reuse but *not* in total budget, and the difference matters for one claim. Batch 8, 16 and 64 were configured for 48 000, 48 000 and 48 640 folds; batch 32 ran to 105 600. Batch 64 peaked at a smoothed pass of 0.605 and never reached 90 % — but it was never given the 55 040 folds that batch 32 needed to get there, so that is a censored observation and not a measurement. The honest statement is that batch 64 does not reach 90 % within 48 640 evaluations. Batch 8 is cheaper at every level up to 70 % and then needs 46 720 to close, so the ordering inverts between 70 % and 90 % — small batches climb fast and converge late, as expected when the GRPO advantage is normalised by the standard deviation of few samples.

What the in-loop chimera pass rate counts. The training-time chimera set is built by rotation: chimera (ρ, i) takes slot j from parent $(i + \rho j) \bmod B$, so at $\rho = 0$ every slot comes from the same parent and the "chimera" is the unrecombined parent. With four rotations, one in four of the chimeras scored inside the loop is a parent, and the levels quoted above are therefore a quarter parental. This does not distort the comparison between arms — the rotation count is four in every arm, so the parental fraction is identical — but it does mean the absolute level is not a pure recombinant pass rate. Every *library* number in this paper is clean: the library evaluation excludes exact parents explicitly, so the retention table and Section 4 are unaffected.

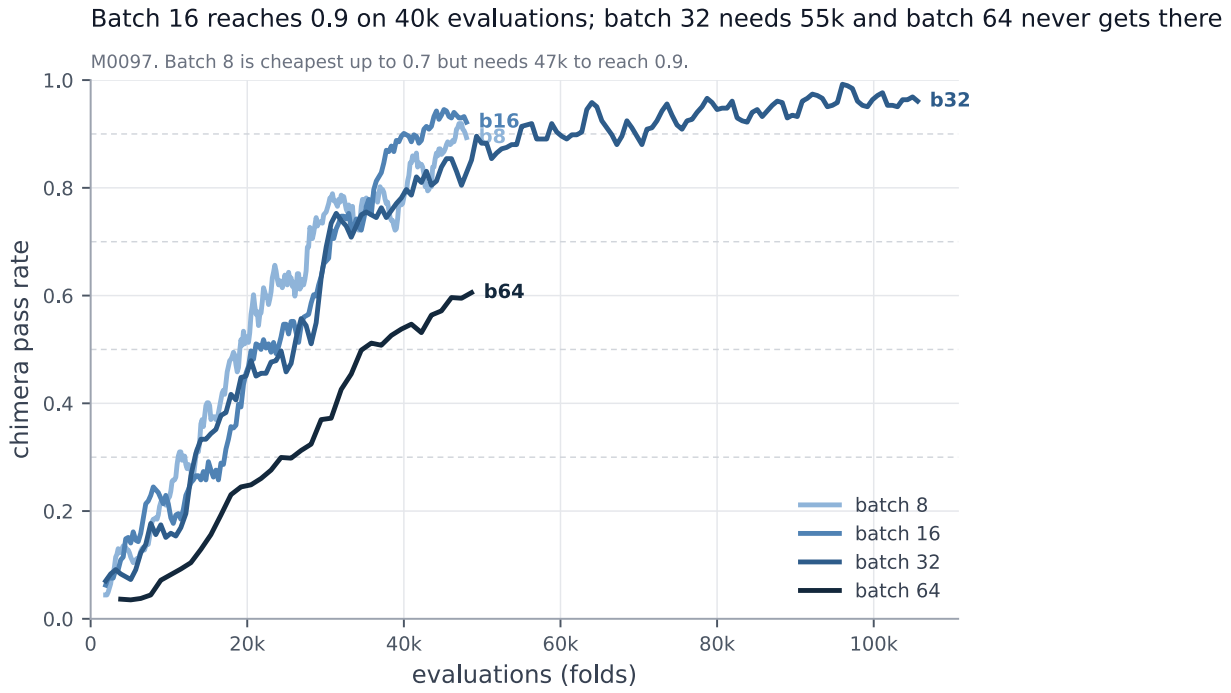


Figure 4: **Evaluation efficiency on M0097.** Chimera pass rate against cumulative structure predictions for four batch sizes at matched sample reuse. Curves are smoothed over a constant *fold* width rather than a constant number of steps: a three-step window spans 480 folds at batch 8 and 1920 at batch 32, which smooths the small-batch arms four times less and lets a noisier curve touch a threshold early by chance. Window width is held at 1920 folds and kept a multiple of three so that the round-robin oracle pool stays balanced.

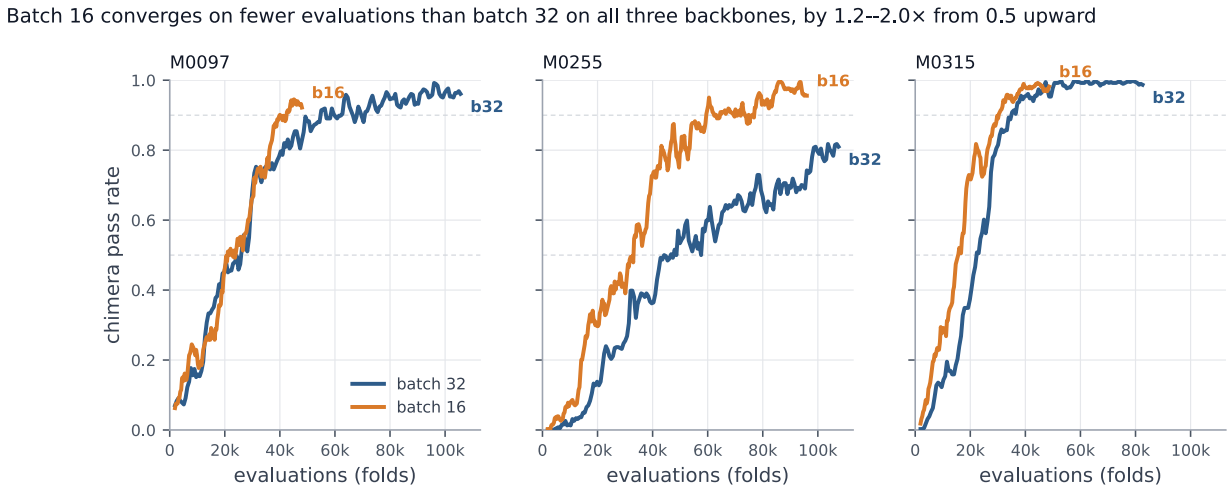


Figure 5: **The ordering replicates across backbones.** Batch 16 against batch 32 on all three AME backbones. The margin is *smallest* on M0097, the backbone the sweep itself was run on. On M0255 — the hardest of the three — batch 16 reaches 90 % at 59 520 folds where batch 32 never reaches it at all within 107 520.

Why this block is here and not in methods. It reads as a hyperparameter note but it is the reason the held-out AF3 probe grid (project ledger, §4 — not reproduced here) and the library result in Section 4 are affordable at all. It also carries one correction worth keeping visible: every batch comparison made before this sweep held `N_updates` fixed at 16, which makes sample reuse scale as 1/batch — 16× at batch 8 against 4× at batch 32. Those comparisons varied batch and reuse together, so the earlier conclusion that batch 8 has a lower ceiling was not a statement about batch size. Pinning reuse is what makes this one a one-factor result.

Still owed: every cell is $n = 1$ in seed, and the figures above are training-level pass rates. The held-out AF3 probe grid agrees with the ordering (batch 16's final held-out pass on M0255 is 0.938 against batch 32's 0.789), but no arm here has been replicated.

4 Results: fragment recombination builds the library

Claim. Designs from the trained policy can be cut at fixed boundaries and recombined across parents, so that $\sum_i |F_i|$ synthesized DNA parts yield $\prod_i |F_i|$ testable variants — and the recombinants keep enough of the parental pass rate that the resulting library is worth building.

Status: scaffold. The computational half is measured; the section is still written as a plan because the blocks below are what the prose has to become. The *physical* half is no longer hypothetical: Section 5 reports a five-fragment library built by exactly this procedure — 128 protein parts spanning 1.1×10^7 recombinants, 3,135 of them assembled and assayed, 67% registering above background at a median of 29 mutations from the parent. That is the one experimental test in this paper of the independence assumption named below.

Why it is the load-bearing section. Section 3 buys a better *distribution*; on its own that is a sampling result. This section is what converts a distribution into something you can physically order: a library assembled from k fragment sets is the product of their sizes, so recombination is what buys library scale that sampling alone cannot. Without it the paper is a benchmark result.

Block 1 — The combinatorial argument, stated precisely

- Set up the notation: L residues split into k fragments at fixed boundaries $[s_i, e_i]$; F_i the set of distinct sequences observed at fragment i across the policy's output; a recombinant is one choice per slot, joined.
- The arithmetic: $\sum_i |F_i|$ parts, $\prod_i |F_i|$ recombinants. With $k = 5$ and $|F_i| = 40$ that is 200 parts and 1.0×10^8 variants. **State the actual numbers used, not this example.**
- Golden Gate assembly is what makes the product physically reachable: one pot, orthogonal overhangs at the boundaries, no per-variant synthesis. Methods carries the adapter design.
- **The assumption the whole section rests on, named up front:** fragments are treated as if they contribute independently to foldability. They do not, exactly — the whole point of an inverse folding model is that residues are coupled. Block 3 measures how badly the approximation fails, and that measurement is the result.

[TODO: fragments — state the actual k, $|F_i|$ and library size used for the AME arm. The PETase numbers ($k = 5$, $|F_i| = 25$–26, 1.1×10^7) are in Section 5 and should be quoted here as the physical instance, not substituted for the computational one.]

Block 2 — Where the boundaries go

- Boundaries are chosen on the structure, not the sequence: cut in loops and between secondary-structure elements, never through the active site.
- **Pre-register the comparison:** boundaries placed on secondary-structure breaks should retain more pass rate than boundaries placed at evenly spaced sequence positions. If they do not, the structural rationale is decorative and should be dropped from the paper rather than defended.

- Report the fixed-residue set — catalytic residues held constant across all fragments, so every recombinant retains the active site by construction.

[TODO: fragments — boundary placement rule + the structural-vs-uniform control]

Block 3 — Do recombinants fold? (the experiment that must be run)

This is the measurement the section exists for. Everything above is setup and everything below is consequence. Design, in the order it should be run:

1. Sample n designs from a trained CLEO policy with no selection applied, on an AME backbone where the parental pass rate is already known and high (M0097 is the obvious choice — it is the backbone every other measurement in the paper uses).
2. Split each at the fragment boundaries. Keep only fragments drawn from *passing* parents, since that is what one would actually order.
3. Draw recombinants uniformly from the product space, excluding any recombinant that reproduces a parent exactly.
4. Fold best-of-5 and score on the identical code path as Section 3. **Same pipeline, same reference, or the comparison to the parental rate is not a comparison.**

The endpoint is retention: recombinant pass rate divided by parental pass rate. Everything else is secondary.

Pre-registered predictions, written before the run:

- Retention is well above the rate for recombining across *random* MPNN samples at matched diversity. If it is not, fragmentation is adding nothing that temperature would not.
- Retention falls monotonically with the number of slots swapped away from a single parent. A recombinant differing at one slot should behave nearly like its parent; one differing at all k should be worst. A flat curve would mean the fragments are effectively independent, which would be a stronger result than we expect and should be reported as such.
- Recombinants reach mutational combinations absent from the parental set — measured as U over the recombinant library against U over the parents at matched design count.

What kills the claim: retention below the point where the library is filterable in practice. If the parental rate is 57 % and recombinants pass at 2 %, the product space is nominal rather than real, and the honest paper reports the fragment library as a screening funnel with a stated hit rate instead of as a 10^8-member library. Decide the threshold now, not after seeing the number.

This experiment has now been run. The rest of this block reports it. The pre-registration above is left in place so the predictions can be read against the outcome rather than reconstructed after it.

Three AME backbones, 256 designs sampled per arm with no selection applied, split at uniform boundaries into $k = 4$ slots, recombined uniformly over the product space with exact parents excluded, and folded best-of-5 through `af3_from_df` — the identical evaluator used for the parental rate and for every library number in this paper, so the comparison is a comparison.

Recombinants fold, and training for recombination is what makes them fold. Under the chimera objective at matched batch the recombinant pass rate is *above* the parental rate on M0097 (85.5% against 84.4%) and on M0255 (97.3% against 94.9%), and 1.5 points below it on M0315 (97.3% against 98.8%, where the parents are nearly saturated). Cutting a passing design into four parts and reassembling it across parents costs at most a point and a half of foldability, and on two of three backbones costs nothing at all — so the product space is real rather than nominal.

The comparison that carries weight is against the seed band, not against a single baseline. The two plain-RL rows per backbone are the same configuration run twice, and they

Table 2: Library evaluation at $n = 256$ parents and $n = 256$ recombinants per cell, $k = 4$ slots, scored by held-out AF3 best-of-5. The two plain-RL rows per backbone are *replicates of one configuration differing only in* `seed`. The bold chimera row is the batch-matched one: $B = 16$, as the plain-RL arms are. The $B = 32$ row is shown because it was the arm reported in earlier drafts, and because the two differ enough to matter. Sample reuse is still unmatched between objectives (4× for chimera, 8× for plain RL); retention is deliberately omitted as a column — see below.

backbone	objective	B	seed	parents	recombinants
M0097	plain RL	16	2	88.7%	68.0%
M0097	plain RL	16	3	44.1%	71.1%
M0097	**chimera**	16	1	84.4%	**85.5%**
M0097	chimera	32	1	94.1%	96.9%
M0255	plain RL	16	2	59.0%	29.3%
M0255	plain RL	16	3	33.6%	30.5%
M0255	**chimera**	16	1	94.9%	**97.3%**
M0255	chimera	32	1	80.5%	82.4%
M0315	plain RL	16	2	89.5%	65.2%
M0315	plain RL	16	3	92.6%	57.0%
M0315	**chimera**	16	1	98.8%	**97.3%**
M0315	chimera	32	1	100.0%	100.0%

bound what a lucky draw can buy. At matched batch the chimera arm clears the full observed band on every backbone: +14.4 points on M0097 (85.5% against 68.0–71.1%), +66.8 points on M0255 (97.3% against 29.3–30.5%), and +32.1 points on M0315 (97.3% against 57.0–65.2%). Measured against each backbone's own seed-to-seed spread in the recombinant rate (3.1, 1.2 and 8.2 points) those margins are 4.6×, 56× and 3.9×.

That is the honest version of a claim earlier drafts made more strongly. The margins quoted previously (+25.8, +52.0, +34.8) came from the $B = 32$ chimera rows, which differ from the plain-RL arms in batch size and sample reuse as well as in objective and oracle. Batch-matching costs M0097 eleven points of margin and gains M0255 fifteen. One caveat survives the correction and should be stated rather than buried: M0097's +14.4 is *below* the 21-point control-pass swing documented between two unseeded runs of a single configuration elsewhere in this work, so on that backbone the margin alone does not clear the widest variance we have measured. M0255 and M0315 clear it comfortably.

Retention is the least reproducible statistic in this table and should not be read as a point estimate. It is a ratio whose numerator is stable and whose denominator is not. Across the plain-RL replicates the recombinant rate moves by 1.2–8.2 points between seeds while the *parental* rate moves by 3.1–44.5 points, so retention swings from 0.767 to 1.611 on M0097 for a configuration that was not changed at all. It has therefore been dropped as a column: the recombinant pass rate is the endpoint, and it is what the table reports.

Retention is also not the same quantity across arms, which is the second reason to stop tabulating it. The recombination step filters parents to those that already pass, so for an arm whose parents are drawn from a pre-filtered pool the denominator is 1.0 by construction, while for the arms here it is the pass rate of the 256 *sampled* designs. Dividing the unselected-MPNN control by the same rule gives retention of 1.53, 7.59 and 1.43 — values that are obviously not foldability ratios, and that show the column was never comparing like with like. This is a general caution about retention as an endpoint, and it revises the framing of the pre-registration above, which named retention as

the endpoint and everything else as secondary.

Scoring the pre-registered predictions. The first holds: recombining across unselected MPNN samples gives a library pass rate of 45.1 % against the chimera arm's 93.1 % pooled over backbones. **The third was ill-posed, and scoring it properly is more informative than the prediction would have been.** It asked whether the recombinant library carries more distinct substitutions than its parents at matched design count. It cannot. Every recombinant is a concatenation of parent fragments, so every (position, residue) pair it carries appears in some parent *at that position*: the recombinant repertoire is a subset of the parental one by construction, and no training objective can change that. Earlier drafts reported the prediction as holding, on a pair of numbers (1,409 against 1,139) that no script on disk produces and that were rarefied at different depths.

Measured as the ratio it actually is, recombination is a mild subset. At matched design count the recombinant library retains 87 % of the parental repertoire at $k = 4$ (0.876, 0.848, 0.880 on M0097, M0255, M0315) and 92 % at $k = 16$ (0.944, 0.913, 0.913), while multiplying the number of variants carrying it by four to thirty orders of magnitude depending on k. That $k = 16$ loses less than $k = 4$, with no overlap between the two ranges, is the expected direction: shorter fragments mean each slot's substitutions recur across more recombinants, so fewer are lost to any particular draw.

The objective does not make the subset gentler — it makes the set being subsetted larger. The preservation ratio is not what the chimera objective improves. Recombining unselected LigandMPNN draws preserves 93 %, marginally the highest of any arm, because a smaller and more redundant parental repertoire has less unique material to lose: that arm retains 836, 584 and 1,222 substitutions out of 854, 668 and 1,294. The chimera arm at $k = 16$ retains 1,095, 1,187 and 1,320 out of 1,160, 1,300 and 1,445 — a slightly smaller fraction of a substantially larger repertoire, and in absolute terms between 1.1 and 2.0 times as many surviving substitutions, at three times the library pass rate. A ratio close to one is cheap to achieve by having little to lose, which is why the absolute count is the quantity that matters here. **The second is not yet measured.** The retention-versus-slots-swapped curve scores recombinants by their Hamming distance in slot space from the nearest parent. That distance is *computable from the scored libraries already on disk* — earlier drafts recorded it as a missing pipeline feature, which was wrong. The real obstacle is the sampling: uniform recombination over the product space at $k = 4$ puts essentially every draw at three slots swapped (249 of 256 on M0097), so the axis has no support at $d = 1, 2, 4$ and the curve cannot be fitted. Measuring it needs a stratified draw at fixed d, not a new metric. It remains the one pre-registered prediction this experiment does not address, but it is cheap rather than blocked.

The chimera objective buys pass rate by narrowing the policy, and cutting finer buys it back. Pointing the library table at the chimera arms makes a cost visible that a comparison against an obsolete arm had hidden. Against plain RL the chimera arms carry fewer distinct substitutions per library and sit closer together in sequence space, at a markedly higher pass rate:

Three things follow. First, the cost is *within* RL: every chimera arm carries more substitutions than the no-RL control (1,154 and 1,201 against 881) at triple its pass rate, so this is a statement about the objective and not about whether function-aligned search moves the frontier. Second, the cost is concentrated in one cell — M0255 at $k = 4$ falls to Hamming 42.4 against plain RL's 73.2, while M0097 and M0315 sit at 68.6 and 78.0 against 72.0 and 84.4 — and M0255 is also the backbone where the fragment-count sweep is non-monotone, so the two may be one instability. Third, $k = 16$ **dominates** $k = 4$ **on both axes**: more substitutions (950 against 895), greater spread (71.2 against 63.0), the same pass rate, and a product space larger by twenty-five orders of magnitude. The diversity given up by cutting into four parts is largely recovered by cutting into sixteen.

Table 3: Recombined libraries at $n = 256$, three backbones, AF3 best-of-5, all arms at batch 16. Substitutions are distinct (position, residue) pairs over *passing* members, raw and rarefied to a matched count of passing members. Hamming is the mean pairwise distance within the passing library. `sampled` recombines unselected LigandMPNN draws; it is the no-RL control.

arm	library pass	substitutions (raw)	(matched n)	Hamming
sampled	30.5 %	881	881	75.3
plain RL	76.7 %	1368	1141	76.5
chimera, $k = 4$	93.4 %	1154	895	63.0
chimera, $k = 16$	92.2 %	1201	950	71.2

Two limits to state plainly when this is written for submission.

(i) The chimera cells are $n = 1$ in seed. Every chimera-objective run in this paper uses `seed: 1`, and the replicate evidence above comes from plain-RL arms trained against RF3 alone rather than the three-oracle pool. A single-oracle policy can overfit that predictor differently on each run, which is a plausible mechanism for the 44.5-point parental swing and suggests the pool arms may be tighter — but no pool arm has ever been run twice, so the error bar on the chimera row is unmeasured. One replicate of the pool configuration is the cheapest thing that would fix this.

(ii) The contrast varies more than one factor. Batch size is matched in the bold rows, but sample reuse is not: the chimera arms reuse each rollout 4× and the plain-RL arms 8×, because both were configured with `N_updates: 16` at different batch sizes. So the contrast varies objective (chimera vs. plain RL), training oracle (pool vs. RF3 alone) and reuse. It is *consistent with* chimera-plus-pool beating plain-RL-plus-RF3; with one chimera seed and three factors moving it does not establish it, and it does not apportion the gain.

One contrast does isolate the oracle, and it is not the grid this passage used to cite — that grid lives in the project ledger, not in Section 3, and it covers only pool-trained arms, so it cannot speak to pool-versus-single at all. The contrast that can is `fragb32` against `fragb32rf3`: both batch 32 (32 rollouts per step, read off the written metrics rather than the configs), both chimera-objective, both self-resumed with no foreign warm start, differing only in training oracle. Over the held-out AF3 probe points their in-loop overstatement differs by 0.046 (−0.005 for the pool over 14 points against +0.041 for RF3 alone over 10) — so "roughly 0.05" survives, but on this comparison rather than the cited one. It points to the objective carrying most of the effect. Both arms are still $n = 1$ in seed, and this table alone cannot separate the three factors.

Block 4 — Relation to manufacturing-aware generative models

Weinstein et al. [1] is the nearest prior art and the section must engage it directly rather than in passing. Their move is to make the generative model manufacturing-aware from the outset — factorized to match what combinatorial DNA chemistry can physically sample — and they reach $\sim 10^{16}$ designs for antibodies, T-cell antigens and Taq polymerase.

The honest framing. They get there first and at larger scale. What is different is the regime, and it is a real difference rather than a hedge:

- **Their objective comes from a natural distribution.** 300 million observed human antibodies define what a good sample looks like, so the model can be constrained to the chemistry from the start without giving anything up. For a de novo enzyme there is no such distribution — the backbone has no natural homologs — so the objective has to be constructed, which is what Section 3 does.
- **Order of operations.** They factorize first and train the factorized model; we train an unfactorized policy for catalytic geometry and factorize afterwards. That makes retention (Block 3) our problem and not theirs — their samples are manufacturable by construction, ours have to be shown to survive the cut.

- **Scale is not the axis we win on and the section should say so plainly.** 10^{16} is out of reach here and it does not need to be reached: a 10^8 library that is enriched for catalytic geometry is a different object from a 10^{16} library that is not.

Do not write this block as a competitive comparison. The two results compose — a manufacturing-aware architecture trained against a catalytic reward is the obvious next thing, and saying so is more useful than claiming a win.

[TODO: fragments — write Block 4 once Block 3 lands; needs the library-size and retention numbers to be concrete about the 10^8 vs 10^{16} point]

5 Results: a recombined PETase library, assayed

Claim. A fragment-recombination library built by exactly the procedure of Section 4 was ordered, assembled and assayed: 3,135 designs, each 18–76 mutations from its parent, measured on a single run of the MoMI esterase assay. Two things follow. The platform produces physically assemblable, expressible, measurable libraries at combinatorial scale — and zero-shot structure prediction, the signal the policy is trained against, is *lightly* predictive of measured activity on this scaffold — $\rho = 0.58$ at best, worth a 2.8× enrichment, beaten by mutation count alone — and **does not transfer to a second serine hydrolase at all**: across 49 shared AF3 metrics the two enzymes' usefulness rankings anti-correlate at −0.604, and all ten of PETase's most strongly *positive* predictors invert in sign. The oracle calibration is a per-target measurement, not a method property, which is the strongest argument in this paper for deciding a library by experiment rather than by score.

Status: drafted. Every number below is on disk and is regenerated by `paper/figures/petase_zeroshot.py` from `figures/data/petase_activity_zeroshot.csv`; the figure script prints the prose's numbers so the two cannot drift. What this section does *not* contain is a closed-loop activity campaign — see "What this does not show" at the end, and the plan in `archive/results_petase.tex`.

The library is the method, physically

The assayed library is a five-fragment Golden Gate recombination library over a 209-residue de novo PETase scaffold with the Ser66/His107/Asp122 triad. Boundaries fall at residues giving fragments of 42, 44, 42, 37 and 44 residues. The curated fragment dictionary carries 25, 26, 26, 26 and 25 variants at the five slots: **128 protein parts spanning 1.1×10^7 recombinants**, and across library generations 1,920 DNA parts were ordered as BsaI-flanked blocks and assembled one-pot from an acoustic-transfer map, one well per recombinant. This is the arithmetic of Section 4 instantiated in DNA: the ratio of orderable parts to reachable variants is roughly 10^5.

What was recovered and assayed is 3,135 designs carrying a median of 29 mutations from the parent (interquartile range 25–38, full range 18–76). **67% register rate above background**, which is the first result: a library recombined across parents at this mutational depth is mostly not dead. That is the assumption Section 4 names as load-bearing – that fragments contribute roughly independently to foldability – and it is the only direct experimental test of it in this paper.

Zero-shot prediction is lightly predictive

Every design was folded by AF3 in two states — the Michaelis complex and a covalent *tetrahedral adduct* — and scored on 52 confidence and geometry metrics. The adduct is not an acyl-enzyme intermediate, although the column is named `acyl_*`: AF3 strips the leaving hydrogens but keeps the leaving oxygens, so the acid –OH survives and the bonded carbon stays 4-coordinate. Both enzymes in the transfer comparison below model the same species, so that comparison is unaffected

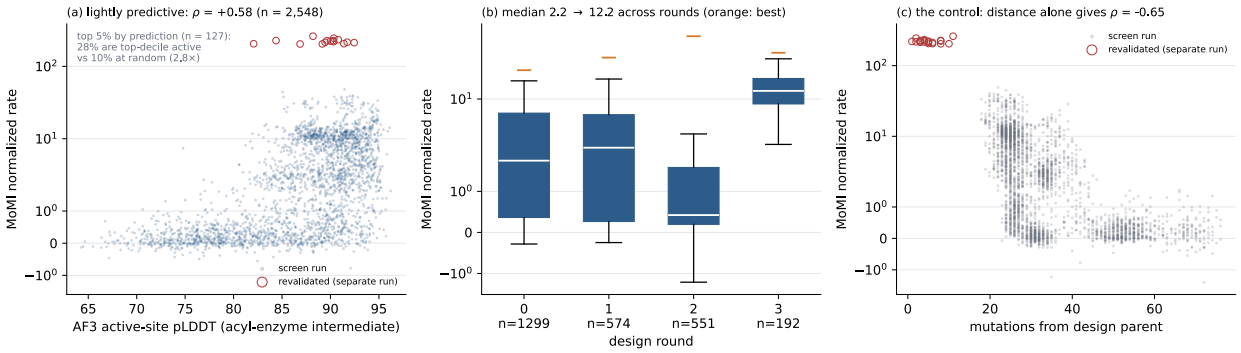


Figure 6: **Structure-prediction confidence against measured activity for 3,135 recombined PETase designs.** (a) The strongest zero-shot signal is local and state-specific: per-atom pLDDT over the active site of the *tetrahedral adduct*. Selecting the top 5% by this metric returns top-decile actives at 2.8× the base rate. (b) Activity by design round, on the single screen run. (c) The control: distance from the design parent ranks activity better than any prediction does. The 20 open red points are a selected set re-assayed in a different run of the same assay and are never pooled into a correlation.

— but the state is the adduct, and calling it the acyl-enzyme intermediate would misstate which chemistry the predictor was scored on. Table 4 gives the ranking against measured activity.

Predictor	What it measures	Spearman ρ	n
Zero-shot, AF3			
acyl His107-NE2 pLDDT	catalytic atom, acyl state	+0.60	2,548
acyl active-site pLDDT	active-site shell, acyl state	+0.58	2,548
acyl PAE Ser66↔His107	triad geometry	−0.47	2,548
substrate interface PAE (min)	protein–ligand placement	−0.43	2,616
substrate pTM	global confidence	+0.36	2,616
substrate ipTM	global interface confidence	+0.32	2,616
Zero-shot, sequence only			
mutations from parent	distance, no structure at all	−0.65	3,135
Supervised, held out (RidgeCV, 200 stratified splits)			
AF3 features		$+0.73 \pm 0.02$	
Rosetta features		$+0.75 \pm 0.01$	
AF3 + Rosetta		$+0.80 \pm 0.01$	

Table 4: **Ranking power against measured activity.** Zero-shot rows are univariate Spearman on the screen run. Global confidence is roughly half as informative as a local, state-specific measurement at the catalytic atoms.

Three readings, in the order that matters.

The useful signal is local and state-specific, not global. pTM and ipTM — the metrics a design pipeline reaches for by default — reach $\rho \approx 0.35$. Per-atom pLDDT at the catalytic histidine of the *tetrahedral adduct* reaches 0.60. The gain comes from scoring the chemically relevant state at the chemically relevant atoms, which is an argument for building the covalent intermediate into the oracle rather than for folding harder.

It is a real signal, and it is light. Selecting the top 5% of the library by active-site pLDDT

returns a top-decile-active design 28% of the time against a 10% base rate: a 2.8× enrichment, falling to 1.9× if 25% of the library is kept. That is worth having when the alternative is random ordering, and it is nowhere near a filter one would trust to replace an assay.

And a cheaper predictor beats it. Mutation count from the parent, which requires no structure prediction at all, ranks activity at $\rho = -0.65$ — better than any AF3 metric in the table (Figure 6c). This is the comparison a zero-shot claim has to clear and it does not clear it. What survives is that the AF3 signal is not merely that confound re-expressed: restricted to the 2,539 designs at least 20 mutations from the parent, active-site pLDDT still gives $\rho = +0.59$. Within single 5-mutation bands it ranges from +0.05 nearest the parent to +0.59 at 30 to 34 mutations, and is indistinguishable from zero in the three most distant bands (−0.01 to +0.09 beyond 50). Prediction adds something to distance; it does not dominate it — but it adds it only in the middle of the range, and the band-wise view is what shows that the pooled +0.59 is not uniform.

. . . and it does not transfer to a second enzyme

The $\rho = 0.60$ above is a property of this scaffold, not of the method. The same two-state fold and the same metric pipeline were run against a second, unrelated serine hydrolase — `super`, a 160-aa de novo design (Ser128/His95/Asp53, 4-methylumbelliferyl butyrate, 2,509 purified designs measured over 2.51 decades). Correlating each metric's *usefulness* on one enzyme against its usefulness on the other asks whether a predictor chosen once can be reused. It cannot.

State	Concept	PETase ρ	`super` ρ	Same sign
acyl	His catalytic-N pLDDT	+0.600	−0.147	no
acyl	active-site pLDDT	+0.587	−0.076	no
acyl	His pLDDT	+0.573	−0.080	no
acyl	Ser pLDDT	+0.542	−0.054	no
acyl	Ser-OG pLDDT	+0.526	−0.075	no
acyl	ligand pLDDT (mean)	+0.424	−0.045	no
substrate	Ser pLDDT	+0.385	−0.179	no
delta	interface PAE (min)	+0.371	−0.177	no
substrate	pTM	+0.348	−0.116	no
delta	His catalytic-N pLDDT	+0.345	−0.049	no
mean of these ten		+0.470	−0.100	0/10
all 49 shared concepts		$\rho_\rho = -0.604$		10/49

Table 5: **The zero-shot correlation does not survive an enzyme swap.** Every one of PETase's ten most strongly positive AF3 predictors inverts on a second serine hydrolase. Ranked by $|\rho|$ instead, one of the top ten keeps its sign — the claim is about the predictors anyone would actually pick, which are the positive ones. ρ_ρ is the Spearman between the two enzymes' per-metric Spearmans across all 49 shared concepts: +1 would mean a predictor can be picked once and reused, 0 that one enzyme says nothing about the other, and a negative value that it actively misleads. Sign agreement of 10/49 is worse than the ∼24 expected by chance. Source: `figures/data/petase_super_af3_transfer.csv`.

Rosetta fails differently and more mildly: $\rho_\rho = +0.035$ over 66 shared concepts with 33/66 sign agreement, which is exactly chance. That is *no* transferable signal rather than an inverted one, a weaker claim than AF3's.

The nuance is worth stating, because it is the one place anything survives. PETase's three strongest Rosetta predictors are all His–Asp distance terms, and all three *keep their sign* on `super`

while collapsing in magnitude: $-0.502 \rightarrow -0.072$, $-0.484 \rightarrow -0.023$, $-0.410 \rightarrow -0.030$, a 7–21× shrink. Catalytic-geometry strain points the same way on both enzymes and nearly stops mattering; it is the only concept family here that does. Rank agreement over all 66 concepts is still nil, so this does not rescue selection — but it is a different failure from AF3's outright inversion, and the direction is not random.

What *does* replicate is a methodological fact about the two predictor families. On both enzymes, Rosetta carries substantial information AF3 cannot reconstruct while the reverse is largely false: predicting Rosetta features from AF3 gives a median cross-validated R^2 of 0.145 on PETase and 0.199 on `super`, against 0.556 and 0.527 in the other direction. The most private features are relax-ensemble *variance* terms, for which a single structure prediction has no analogue. Same asymmetry, same magnitude, both enzymes.

This is the wet-lab counterpart of the transfer gap in Section 3. There, a policy trained inside a structure predictor scored higher under that predictor than under a held-out one. Here, a predictor that ranks function on the target it was calibrated against anti-ranks it on the next target. Both say the same thing: a structure predictor's agreement with the quantity you care about is a property of the specific system, re-earned per target, never inherited.

That conclusion is an argument *for* the library, not against it. If zero-shot oracle scores could be trusted across targets, the right strategy would be to rank a large virtual set and order the top of it. Because they cannot, the ordering has to be decided by something the oracle does not control — which is what a combinatorially compatible library at a 67% register rate buys: enough physically assemblable, measurable diversity that the experiment, not the predictor, does the selecting.

Searching the library moves activity

Across four design rounds on the single screen run, median activity rises from 2.2 in round 0 to 12.2 in round 3, and the best single recombinant from 20.5 to 31.7 over the same span — with the highest of all four, 47.8, in round 2 (Figure 6b). The progression is not monotone — round 2 has the *lowest* median of the four and the *best* single design, which is what a deliberately broadened library looks like.

A separate, selected set of 20 designs was re-assayed and scored 204–261, roughly 100× the screen median. **Those numbers are from a different run of the same assay** and the two runs share no normalizing control in the record we have, so this paper does not quote a fold-improvement that crosses that line. The 20 also sit 1–11 mutations from the parent, not out in the recombined library, so they are evidence about a local optimization around a good design — not about what recombination reached.

What this does not show

- **No wild-type comparator.** The parent sequence was never assayed in this table, so no fold-over-parent number can be computed from it, and none is quoted.
- **Activity falls with distance.** The most active designs in the record are the closest to the parent. Recombination at 20–80 mutations buys reachable diversity and a 67% register rate; on this scaffold and this assay it did not buy a better enzyme than careful local work did. Section 3's claim is about the diversity/fidelity frontier, and that is the claim this section supports.
- **The oracle calibration does not generalise, and that is measured, not assumed.** A second serine hydrolase inverts every one of PETase's ten best AF3 predictors (Table 5). So this section calibrates the oracle *on this target* and establishes physical realizability; it does not license zero-shot selection on a new target, and no number here should be carried to one. The AME results carry the generality claim.

- **Two enzymes is not a trend.** $\rho_\rho = -0.604$ is one pairwise comparison. It is enough to refuse the claim that the correlation transfers; it is not enough to predict the sign or size of the failure on a third target.
- **The closed loop is not here.** The surrogate-in-the-reward campaign is planned in `archive/results_petase.tex` and has no data in this repository.

6 Discussion

Science research articles end with a short discussion / outlook rather than a Conclusion section. Keep this tight (∼250–400 words). Write last, after the abstract. The four paragraphs below mirror the three-move spine plus limits.

P-A — The central takeaway

- The active-variant region for a de novo enzyme is farther from the starting design than SSM can reach and more specific than temperature sampling can hit. Both halves of that sentence are measured in this paper, and neither is measured by prior work.
- The resolution is not a better sampler or a bigger library on its own. It is a sampler trained against function, factorized into parts that can be physically assembled, and closed against measured activity. Each piece is necessary; state why.

[TODO: discussion p1 — central takeaway]

P-B — What each move established

- **Frontier.** A KL-anchored RL policy occupies a region of the diversity/fidelity plane that no temperature reaches. The anchor is the mechanism, not a detail — unanchored, the policy collapses onto a narrow solution with worse per-design coverage than the baseline.
- **Fragments.** Recombination converts that distribution into a library exponentially larger than the parts ordered, at a measured cost in pass rate.
- **PETase.** Three rounds, each designed on the last round's data, ending above characterized natural enzymes on the same probe.

[TODO: discussion p2 — what each move proves]

P-C — Honest limitations

- **Bulk PET.** Probe activity is not plastic degradation. Do not let this paragraph soften.
- **Run-to-run variance.** Single-run comparisons of pass rate on this benchmark are not interpretable at the effect sizes usually of interest; two draws of one config moved 21 points. Every claim retained in this paper either clears that width or is replicated. *Cite the notebook's E17/E19/E21 record for the measurement.*
- **The reward and the evaluation share a predictor.** Motif RMSD under AF3 appears on both sides of the frontier result. A held-out oracle remains the outstanding control, and the paper should say so rather than wait to be asked.
- **Selection during training does not help.** Five experiments looked for a rule that beats random selection of rollouts and none found one. Worth one sentence as a negative result others need not repeat.
- **Cold start.** The loop needs a first library with measurable spread.

[TODO: discussion p3 — limitations]

P-D — Outlook

- The obvious composition: a manufacturing-aware architecture [1] trained against a catalytic reward, rather than factorizing after the fact as we do here. That gets the scale of the former and the objective of the latter, and it is the next experiment rather than a distant prospect.
- More rounds, and rewards that mix computational and measured signal continuously rather than once per library.
- Other chemistries and substrate classes, including bulk polymers.

[TODO: discussion p4 — outlook]

7 Methods: shared pipeline

Shared methodology for the computational results. Per-section specifics live in the following subsections. Do not write this out in detail before the Results numbers exist — Methods text written against placeholder numbers gets rewritten.

Block 1 — RL fine-tuning of an MPNN designer with GRPO

- Reference `cleo.design.train_policy` and `cleo.design.utils.grpo`.
- Reward pipeline composition (`UniversalReward`, ordered metric steps, normalized weighted aggregation).
- Cite GRPO [11], ProteinMPNN [7] and LigandMPNN [8].
- **State which base model each result uses.** The AME benchmark arms are LigandMPNN, since those targets have a ligand in the backbone; the PETase campaign is ProteinMPNN. The algorithm is identical across both — only the base policy differs — but a reader cannot be left to infer that from context.

[TODO: methods overview — GRPO + reward pipeline]

Block 2 — Fragment-based combinatorial libraries

- Sampling, fragment splitting, combinatorial recombination.
- DNA assembly via Golden Gate adapters.

[TODO: methods overview — fragment library construction]

Block 3 — Surrogate models for closed-loop optimization

- MLP ensemble with Gaussian NLL.
- Acquisition function (BatchUCB + diversity).
- Used as a reward step inside the RL pipeline.

[TODO: methods overview — surrogate ensemble + acquisition]

Block 4 — Batch-consensus mutation-diversity reward

- Per-position modal AA across the batch (ties multi-modal).
- Two metrics: `consensus_divergence` (count) and `fractional_divergence` ($1/k$ weighted, penalizes secondary-mode collapse).
- Reference `cleo.design.utils.mutation_diversity`.
- Contrast with the embedding-level diversity regularizer of ProteinZero [13] and diversity-regularized DPO [14]: ours acts in mutation space against a named reference sequence, theirs in embedding space.

[TODO: methods overview — diversity reward definition]

References

[1] Eli N. Weinstein, Mattia G. Gollub, Andrei Slabodkin, Kerry Dobbs, Xiao-Bing Cui, Cameron L. Gardner, Ryan J. Grant, Kristina Gurung, Amira Bailey, Alan N. Amin, George M. Church, and Elizabeth B. Wood. Manufacturing-aware generative models enable petascale synthesis of designed dna. *Nature Biotechnology*, March 2026. ISSN 1546-1696. doi: 10.1038/s41587-026-03020-8. URL http://dx.doi.org/10.1038/s41587-026-03020-8.

[2] Joseph L. Watson, David Juergens, Nathaniel R. Bennett, Brian L. Trippe, Jason Yim, Helen E. Eisenach, Woody Ahern, Andrew J. Borst, Robert J. Ragotte, Lukas F. Milles, Basile I. M. Wicky, Nikita Hanikel, Samuel J. Pellock, Alexis Courbet, William Sheffler, Jue Wang, Preetham Venkatesh, Isaac Sappington, Susana Vázquez Torres, Anna Lauko, Valentin De Bortoli, Emile Mathieu, Sergey Ovchinnikov, Regina Barzilay, Tommi S. Jaakkola, Frank DiMaio, Minkyung Baek, and David Baker. De novo design of protein structure and function with rfdiffusion. *Nature*, 620(7976):1089–1100, July 2023. ISSN 1476-4687. doi: 10.1038/s41586-023-06415-8. URL http://dx.doi.org/10.1038/s41586-023-06415-8.

[3] Woody Ahern, Jason Yim, Doug Tischer, Saman Salike, Seth M. Woodbury, Donghyo Kim, Indrek Kalvet, Yakov Kipnis, Brian Coventry, Han Raut Altae-Tran, Magnus S. Bauer, Regina Barzilay, Tommi S. Jaakkola, Rohith Krishna, and David Baker. Atom-level enzyme active site scaffolding using rfdiffusion2. *Nature Methods*, 23(1):96–105, December 2025. ISSN 1548-7105. doi: 10.1038/s41592-025-02975-x. URL http://dx.doi.org/10.1038/s41592-025-02975-x. Print issue Nature Methods 23(1), January 2026.

[4] Anna Lauko, Samuel J. Pellock, Kiera H. Sumida, Ivan Anishchenko, David Juergens, Woody Ahern, Jihun Jeung, Alexander F. Shida, Andrew Hunt, Indrek Kalvet, Christoffer Norn, Ian R. Humphreys, Cooper Jamieson, Rohith Krishna, Yakov Kipnis, Alex Kang, Evans Brackenbrough, Asim K. Bera, Banumathi Sankaran, K. N. Houk, and David Baker. Computational design of serine hydrolases. *Science*, 388(6744), April 2025. ISSN 1095-9203. doi: 10.1126/science.adu2454. URL http://dx.doi.org/10.1126/science.adu2454.

[5] Donghyo Kim, Seth M. Woodbury, Woody Ahern, Doug Tischer, Alex Kang, Emily Joyce, Asim K. Bera, Nikita Hanikel, Saman Salike, Rohith Krishna, Jason Yim, Samuel J. Pellock, Anna Lauko, Indrek Kalvet, Donald Hilvert, and David Baker. Computational design of metallohydrolases. *Nature*, 649(8095):246–253, December 2025. ISSN 1476-4687. doi: 10.1038/s41586-025-09746-w. URL http://dx.doi.org/10.1038/s41586-025-09746-w.

[6] Markus Braun, Adrian Tripp, Morakot Chakatok, Sigrid Kaltenbrunner, Celina Fischer, David Stoll, Aleksandar Bijelic, Wael Elaily, Massimo G. Totaro, Melanie Moser, Shlomo Y. Hoch, Horst Lechner, Federico Rossi, Matteo Aleotti, Mélanie Hall, and Gustav Oberdorfer. Computational enzyme design by catalytic motif scaffolding. *Nature*, 649(8095): 237–245, December 2025. ISSN 1476-4687. doi: 10.1038/s41586-025-09747-9. URL http://dx.doi.org/10.1038/s41586-025-09747-9.

[7] J. Dauparas, I. Anishchenko, N. Bennett, H. Bai, R. J. Ragotte, L. F. Milles, B. I. M. Wicky, A. Courbet, R. J. de Haas, N. Bethel, P. J. Y. Leung, T. F. Huddy, S. Pellock, D. Tischer, F. Chan, B. Koepnick, H. Nguyen, A. Kang, B. Sankaran, A. K. Bera, N. P. King, and D. Baker. Robust deep learning–based protein sequence design using proteinmpnn. *Science*, 378(6615):49–56, October 2022. ISSN 1095-9203. doi: 10.1126/science.add2187. URL http://dx.doi.org/10.1126/science.add2187.

[8] Justas Dauparas, Gyu Rie Lee, Robert Pecoraro, Linna An, Ivan Anishchenko, Cameron Glasscock, and David Baker. Atomic context-conditioned protein sequence design using ligandmpnn. *Nature Methods*, 22(4):717–723, March 2025. ISSN 1548-7105. doi: 10.1038/s41592-025-02626-1. URL http://dx.doi.org/10.1038/s41592-025-02626-1.

[9] Josh Abramson, Jonas Adler, Jack Dunger, Richard Evans, Tim Green, Alexander Pritzel, Olaf Ronneberger, Lindsay Willmore, Andrew J. Ballard, Joshua Bambrick, Sebastian W. Bodenstein, David A. Evans, Chia-Chun Hung, Michael O'Neill, David Reiman, Kathryn Tunyasuvunakool, Zachary Wu, and John M. Jumper. Accurate structure prediction of biomolecular interactions with alphafold 3. *Nature*, 630(8016):493–500, May 2024. ISSN 1476-4687. doi: 10.1038/s41586-024-07487-w. URL http://dx.doi.org/10.1038/s41586-024-07487-w. Author list truncated here; see DOI for the complete list.

[10] Jeremy Wohlwend, Gabriele Corso, Saro Passaro, Noah Getz, Mateo Reveiz, Ken Leidal, Wojtek Swiderski, Liam Atkinson, Tally Portnoi, Itamar Chinn, Jacob Silterra, Tommi Jaakkola, and Regina Barzilay. Boltz-1: Democratizing biomolecular interaction modeling. November 2024. doi: 10.1101/2024.11.19.624167. URL http://dx.doi.org/10.1101/2024.11.19.624167. bioRxiv preprint.

[11] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models, 2024. URL https://arxiv.org/abs/2402.03300. Introduces Group Relative Policy Optimization (GRPO).

[12] Chloe Hsu, Robert Verkuil, Jason Liu, Zeming Lin, Brian Hie, Tom Sercu, Adam Lerer, and Alexander Rives. Learning inverse folding from millions of predicted structures. April 2022. doi: 10.1101/2022.04.10.487779. URL http://dx.doi.org/10.1101/2022.04.10.487779. bioRxiv preprint.

[13] Ziwen Wang, Jiajun Fan, Ruihan Guo, Thao Nguyen, Heng Ji, and Ge Liu. Proteinzero: Self-improving protein generation via online reinforcement learning, 2025. URL https://arxiv.org/abs/2506.07459.

[14] Ryan Park, Darren J. Hsu, C. Brian Roland, Maria Korshunova, Chen Tessler, Shie Mannor, Olivia Viessmann, and Bruno Trentini. Improving inverse folding for peptide design with diversity-regularized direct preference optimization, 2024. URL https://arxiv.org/abs/2410.19471.

[15] Alexi Gladstone, Heng Ji, and Yilun Du. Explorative modeling: Unlocking a third pretraining axis and end-to-end generation, 2026. URL https://arxiv.org/abs/2607.27372.

[16] Kerr Ding, Michael Chin, Yunlong Zhao, Wei Huang, Binh Khanh Mai, Huanan Wang, Peng Liu, Yang Yang, and Yunan Luo. Machine learning-guided co-optimization of fitness and diversity facilitates combinatorial library design in enzyme engineering. *Nature Communications*, 15(1), July 2024. ISSN 2041-1723. doi: 10.1038/s41467-024-50698-y. URL http://dx.doi.org/10.1038/s41467-024-50698-y.

[17] Jason Yang, Ravi G. Lal, James C. Bowden, Raul Astudillo, Mikhail A. Hameedi, Sukhvinder Kaur, Matthew Hill, Yisong Yue, and Frances H. Arnold. Active learning-assisted directed evolution. *Nature Communications*, 16(1), January 2025. ISSN 2041-1723. doi: 10.1038/s41467-025-55987-8. URL http://dx.doi.org/10.1038/s41467-025-55987-8.

[18] Grant M. Landwehr, Jonathan W. Bogart, Carol Magalhaes, Eric G. Hammarlund, Ashty S. Karim, and Michael C. Jewett. Accelerated enzyme engineering by machine-learning guided cell-free expression. *Nature Communications*, 16(1), January 2025. ISSN 2041-1723. doi: 10.1038/s41467-024-55399-0. URL http://dx.doi.org/10.1038/s41467-024-55399-0.